Aditya Agarwal

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Education

Massachusetts Institute of Technology — Cambridge, MA
Ph.D. in Electrical Engineering and Computer Science, GPA: 5.0/5.0 — Sep 2023 – Present

International Institute of Information Technology Hyderabad — Hyderabad, India
M.S. by Research in Computer Science and Engineering, GPA: 9.83/10.0 (Top 2%) — Aug 2021 – May 2023

PES University (formerly PES Institute of Technology) — Bangalore, India
B.E. in Computer Science and Engineering, GPA: 9.31/10.0 (Top 10%) — Aug 2013 – May 2017

Kendriya Vidyalaya — New Delhi, India
Class XII AISSCE: 94.2% (Top 1.5% in India); Class X AISSE: 10.0/10.0 (Top 1%) — Apr 2010 – Mar 2013

Industry Experience

Sycamore — Palo Alto, CA
AI Research Intern — Jun 2026 – Aug 2026

– Built agent-driven embodied reasoning and planning harnesses for drone-based warehouse inspection and released a warehouse-scene aerial-inspection benchmark in NVIDIA Isaac Sim for evaluating frontier models on task planning, decision-making, and mission-control code generation.

Autodesk Robotics — San Francisco, CA
Research Intern, Robotics Perception — May 2025 – Aug 2025

– Developed *Rivet6D*, a feed-forward 6-DoF pose-estimation model for automated robotic part assembly in structured, real-world settings. Devised a simulation strategy to generate realistic, physically plausible training data, and designed the model to disentangle rotation and translation by combining foundation models with equivariant point-cloud networks. Conducted this work with Yotto Koga and Sachin Chitta in Autodesk Robotics' Adaptive Product Assembly group.

Microsoft India R&D — Hyderabad, India
Data Scientist II, Bing Search Technology Center India — Jan 2018 – Mar 2021

– **Related QnA (People Also Ask), Bing STCI:** Expanded PAA to millions of Bing users across 100+ languages and 200+ markets by developing deep-learning and universalization techniques; improved answer coverage and relevance across English, French, and German markets.
– **Azure Health Data Workbench, Azure Global Engineering:** Enabled secure, analytics-ready migration of medical records to Azure by automating preprocessing, de-identification, and ingestion from on-premise healthcare systems into queryable storage; developed analysis and visualization tools to surface insights.
– **Project Sangam, Azure Global Engineering:** Scaled nationwide sanitation training by owning deployment monitoring, API automation, certificate rewards, and tooling for Project Sangam and onboarding the Swachh Bharat e-Learning Portal; supported 1M+ enrollments and trained 110,000+ municipal functionaries across 4,000+ cities through Microsoft India's partnership with the MoHUA, Govt of India. [launch]

Microsoft Research — Bangalore, India
Research Intern — Nov 2015 – May 2016; Nov 2016 – May 2017

– Built *Second Opinion*, a healthcare platform connecting patients with specialists across Indian hospitals, and developed ML models to predict diseases from medical histories and lab results with the Microsoft Intelligent Network for Eyecare (MINE) and Microsoft India Development Center.
– Developed retrieval workflows and visualizations for Massively Empowered Classrooms (MEC) and helped deploy the blended-learning platform as the Mauritius Institute of Education's Virtual Campus; the resulting MSR–MIE collaboration culminated in its official launch by the Mauritius Ministry of Education. [launch]

Academic Experience

MIT Computer Science and Artificial Intelligence Laboratory — Cambridge, MA
Graduate Research Assistant, Learning and Intelligent Systems Group — Sep 2023 – Present

- **3D world representations for robotic manipulation:** Advised by Professors Leslie Pack Kaelbling and Tomás Lozano-Pérez. Built *SceneComplete*, which reconstructs complete, registered object meshes from a single RGB-D view of a cluttered scene, enabling collision-aware and dexterous grasping (RA-L 2025); developed *SCOUT*, which routes each observation to the reconstruction method that maximizes task utility under arbitrary cost constraints, reducing utility regret in downstream manipulation (IROS 2026).
- **Embodied reasoning, decision-making, and planning:** Built systems for long-horizon mobile manipulation, including a semantic-geometric 3D belief for interactive object search that selects perception, navigation, and interaction actions, achieving 70% success versus 40% for geometry-only search and 35% for LLM planning; and *Feijoa*, an LLM agent that constructs task-relevant representations just in time from raw RGB-D observations by composing perception, spatial reasoning, and feasibility-aware skills (both under review).
- **Scalable, representation-driven manipulation learning:** Developed *PROXY*, a pipeline that learns object-centric manipulation policies from a single RGB-D image and task description with zero human or robot demonstrations by generating and filtering synthetic video demonstrations; matched policies trained on human demonstrations across four real-world tasks and two robot embodiments (RSS 2026 Diff4RL and RoboData Workshops).
- **Research mentorship:** Advised the theses of four graduated MIT M.Eng. students and mentored five MIT undergraduate researchers across 3D representation learning, generative modeling, and robotic manipulation.

Université de Montréal & Mila — Montreal, Canada
Visiting Researcher, Robotics and Embodied AI Lab — May 2023 – Aug 2023

- Advised by Professors Liam Paull and Florian Shkurti. Integrated the open-vocabulary 3D scene representation *ConceptGraphs* with a robotic tabletop manipulator. Resulting paper published at ICRA 2024.

IIIT Hyderabad — Hyderabad, India
Research Fellow — Oct 2019 – Jul 2020; Feb 2021 – May 2023

- **Robotics Research Center:** Advised by Professors Madhava Krishna and Srinath Sridhar.
- Advised undergraduate and graduate students on generalizable motion planning by learning to generate collision-free, kinematically valid trajectories guided by an ensemble of cost functions. (ICRA 2024)
- Learned NeRF priors for 512-pixel, high-resolution representations across categories (NeurIPS 2023), and worked on 3D point-cloud shape completion in arbitrary poses for improved grasp estimation (ICRA 2023).
- Led four undergraduate researchers studying synergies between pushing and grasping for tabletop manipulation (CASE 2023); also developed an end-to-end rearrangement and planning pipeline that placed third in the ICRA 2022 Robotic Grasping and Manipulation Competition.
- **Centre for Visual Information Technology:** Advised by Professors C. V. Jawahar and Vinay Namboodiri.
- Proposed video-to-video face swapping for movie scenes that preserves source identity and expression while retaining target pose and background. Outperformed existing SOTA methods. (WACV 2023).
- Developed a scalable lipreading-training framework using talking-head generation and text-to-speech models, supporting arbitrary accents and languages without vocabulary constraints (WACV 2023).
- Proposed an implicit neural video representation whose hypernetwork learns a prior over videos, achieving SOTA results across several generative video tasks (TMLR 2022).
- Developed a personalized lipreading model for an ALS patient using data augmentation and domain adaptation (BMVC 2021).
- **Language Technologies Research Center:** Advised by Professor Anil Kumar Vuppala.
- Built REED, a convolutional multilingual acoustic model operating on raw speech signals, and achieved SOTA performance on three low-resource Indic languages (SLT 2021).

University of Calgary — Calgary, Canada
MITACS Research Intern — May 2016 – Aug 2016

- Advised by Professor Mike Smith. Developed a low-cost smartphone application to capture and analyze low-frequency audio associated with the Ranchlands Hum using digital signal processing.
- Developed techniques for large-scale calibration of Android device microphones [code].

Research Mentorship

Graduate Researchers

- **Akash Anand**, MIT M.Eng. → Jane Street (Sep 25 – May 26) [website]
  M.Eng. thesis: Which Reconstruction Model Should a Robot Use? Routing Image-to-3D Models for Cost-Aware Robotic Manipulation [thesis]
- **Lucy Cai**, MIT M.Eng. → Stealth Robotics Startup (Sep 25 – May 26) [website]
  M.Eng. thesis: Belief-driven Interactive Object Search in Partially Observable Settings [thesis]
- **Maggie Yao**, MIT M.Eng. → Citadel (Sep 25 – May 26) [website]
  M.Eng. thesis: Generative Video Demonstrations for Object-Centric Robotic Manipulation Policies [thesis]
- **Eugenia Feng**, MIT M.Eng. → Databricks software engineer (Jan 24 – Dec 25) [website]
  M.Eng. thesis: Exploring the Role of Foundation Models for Training Generalist Robot Learning Policies [thesis]

Undergraduate & Visiting Researchers

- **Tiffany Zhang**, MIT sophomore (Feb 26 – Present) [website]
- **Kelly Lu**, MIT research mentee → Decagon (Sep 25 – May 26) [website]
  Research project: Knowing Where to Look: Interactive Object Search via Spatial-Semantic Fusion
- **Benjamin Soria**, MIT senior → Stealth Robotics Startup CTO (Jun 24 – Dec 25) [website]
- **Kallol Saha**, IIIT-H visiting researcher → CMU Ph.D. in Robotics (Jan 23 – Sep 23) [website]
- **Gaurav Singh**, IIIT-H senior → Brown University Ph.D. (Jan 22 – Feb 25) [website]
- **Shankara Narayanan V**, IIIT-H visiting researcher → Stealth Robotics Startup CRO (Oct 21 – May 22) [website]
- **Vishal Reddy Mandadi**, IIIT-H senior → CMU Masters in Robotics (Oct 21 – Sep 22) [website]

Teaching

- 6.4212/6.4210, Robotic Manipulation — Fall 2026
- Teaching Assistant, CSEDU-ML Workshop (IIIT-H, IIT-H, and IIT-D) — 2021–2022

Publications

Journal Publications

1. **A. Agarwal**, G. Singh, B. Sen, T. Lozano-Pérez, and L. P. Kaelbling. "SceneComplete: Open-World 3D Scene Completion in Complex Real-World Environments for Robot Manipulation," *IEEE Robotics and Automation Letters (RA-L)*, 2025. Oral talk at ICRA 2026.

2. B. Sen*, **A. Agarwal***, V. Namboodiri, and C. V. Jawahar. "INR-V: A Continuous Representation Space for Video-Based Generative Tasks," *Transactions on Machine Learning Research (TMLR)*, 2022. [OpenReview] [website]

Conference Publications

1. A. Anand, **A. Agarwal**, and L. P. Kaelbling. "Which Reconstruction Model Should a Robot Use? Routing Image-to-3D Models for Cost-Aware Robotic Manipulation," *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026. Spotlight talk at the ICRA 2026 Rigorous Perception Workshop.

2. Q. Gu*, A. Kuwajerwala*, S. Morin*, K. M. Jatavallabhula*, B. Sen, **A. Agarwal**, K. Ellis, C. M. de Melo, C. Rivera, W. Paul, R. Chellappa, C. Gan, J. B. Tenenbaum, A. Torralba, F. Shkurti, and L. Paull. "ConceptGraphs: Open-Vocabulary 3D Scene Graphs for Perception and Planning," *IEEE International Conference on Robotics and Automation (ICRA)*, 2024; CoRL PRL, TGR, LangRob, and Deployable Workshops, 2023.

3. K. Saha*, V. Mandadi*, J. Reddy*, A. Srikanth, **A. Agarwal**, B. Sen, A. Singh, and M. Krishna. "EDMP: Ensemble-of-Costs-Guided Diffusion for Motion Planning," *IEEE International Conference on Robotics and Automation (ICRA)*, 2024; CoRL LangRob and TGR Workshops, 2023.

4. B. Sen*, G. Singh*, **A. Agarwal***, R. Agaram, M. Krishna, and S. Sridhar. "HyP-NeRF: Learning Improved NeRF Priors Using a Hypernetwork," *NeurIPS*, 2023.

5. V. R. Mandadi, K. Saha, D. Guhathakurta, M. N. Qureshi, **A. Agarwal**, B. Sen, D. Das, B. Bhowmick, A. K. Singh, and M. Krishna. "Disentangling Planning and Control for Non-Prehensile Tabletop Manipulation," *IEEE International Conference on Automation Science and Engineering (CASE)*, 2023.
6. B. Sen*, **A. Agarwal***, G. Singh*, B. Bhowmick, S. Sridhar, and M. Krishna. "SCARP: 3D Shape Completion in Arbitrary Poses for Improved Grasping," *IEEE International Conference on Robotics and Automation (ICRA)*, 2023; RSS Robot Representations Workshop, 2023. [website]
7. **A. Agarwal***, B. Sen*, R. Mukhopadhyay, V. Namboodiri, and C. V. Jawahar. "Towards MOOCs for Lip Reading: Using Synthetic Talking Heads to Train Humans in Lipreading at Scale," *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2023. [website]
8. **A. Agarwal***, B. Sen*, R. Mukhopadhyay, V. Namboodiri, and C. V. Jawahar. "FaceOff: A Video-to-Video Face Swapping System," *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2023. [website]
9. **A. Agarwal***, B. Sen*, S. N. V.*, V. R. Mandadi*, B. Bhowmick, and K. M. Krishna. "Approaches and Challenges in Robotic Perception for Table-Top Rearrangement and Planning," *Robotic Grasping and Manipulation Competition, ICRA*, 2022. [paper]
10. B. Sen*, **A. Agarwal***, R. Mukhopadhyay, V. Namboodiri, and C. V. Jawahar. "Personalized One-Shot Lipreading for an ALS Patient," *British Machine Vision Conference (BMVC)*, 2021. [paper]
11. B. Sen*, **A. Agarwal***, M. S. Ganesh, and A. K. Vuppala. "REED: An Approach Towards Quickly Bootstrapping Multilingual Acoustic Models," *Spoken Language Technology Workshop (SLT)*, 2021. [paper] [presentation]

Workshop Publications

1. M. H. Yao, S. Chintalapudi, A. Anand, M. Tippur, T. Lozano-Pérez, L. P. Kaelbling, and **A. Agarwal**. "PROXY: Learning Object-Centric Manipulation Policies from a Single RGB-D Image," *RSS Diff4RL and RoboData Workshops*, 2026.
2. B. Sen, M. Wang, N. Thakur, **A. Agarwal**, and P. Agrawal. "Learning to Look Around: Enhancing Teleoperation and Learning with a Human-Like Actuated Neck," *CoRL WCBM Workshop*, 2024. Spotlight presentation.

Preprints

1. M. H. Yao, S. Chintalapudi, A. Anand, M. Tippur, T. Lozano-Pérez, L. P. Kaelbling, and **A. Agarwal**. "PROXY: Learning Object-Centric Manipulation Policies from a Single RGB-D Image," 2026.
2. **A. Agarwal***, L. Cai*, K. Lu, T. Manderson, N. Roy, T. Lozano-Pérez, and L. P. Kaelbling. "Knowing Where to Look: Interactive Object Search via Spatial-Semantic Fusion," 2026.
3. **A. Agarwal**[*], W. Shen[*], S. Chintalapudi[†], L. Cai[†], N. Kumar, A. Curtis, T. Lozano-Pérez, and L. P. Kaelbling. "Language Agents that Build Just-in-Time Representations for Mobile Manipulation," 2026.

Patents

1. C. V. Jawahar, **A. Agarwal**, B. Sen, R. Mukhopadhyay, and V. Namboodiri. "Rethinking Approaches to Training Humans with Hearing Loss in Lipreading," provisional U.S. patent, 2023.

[*] *indicates equal first-author contribution;* [†] *indicates equal second-author contribution.* Full publication list at Google Scholar; project details at skymanaditya1.github.io.

Academic Service

Conference Reviewing

- Outstanding Reviewer, CVPR 2025.
- Reviewer for CoRL 2026, NeurIPS 2024, CVPR 2024, ICRA 2023–2024, IROS 2023, SIGGRAPH Asia 2023, and ICLR Workshops 2023.

Leadership and Outreach

- MIT EECS GAAP mentor for the FY2024 cohort and EECS GSA Executive Member.
- Coordinator for the fifth and sixth Summer Schools on AI at CVIT, IIIT Hyderabad, 2021–2022.

Honors and Awards

- Outstanding Reviewer, CVPR 2025, Nashville, Tennessee [link].
- NeurIPS Scholar Award ($1,700) to attend NeurIPS 2023 in New Orleans, Louisiana.
- IEEE Robotics and Automation Society travel grant ($2,250) to attend ICRA 2023 in London, UK.
- Third place ($1,000 prize), ICRA 2022 Robotic Tabletop Manipulation Competition [link].
- Travel grant (¥30,000) to attend ICMPC17-APSCOM7 in Tokyo, Japan.
- Selected among 6,000+ Microsoft employees for the company's campus-hiring video, 2020 [link].
- Delight Your Customer Award, Azure Global Engineering, Microsoft, 2018.
- Winner, VMware Global Relay Open Source Borathon, 2017 [link].
- Academic Distinction Award, PES University, 2013–2017.
- MITACS Globalink Award for a fully funded research internship at the University of Calgary, 2016.
- Co-founder, AndroidLabs, the official Android community of PES University, 2015.
- Best Application Award, Ayana'15 24-hour hackathon, 2015.
- CBSE Certificate of Excellence for placing in the top 0.1% of successful AISSCE Computer Science candidates, 2012.
- Recipient of the INSPIRE Scholarship (deferred) for placing in the top 1% of successful AISSCE 2013 candidates.
- Indian delegate, JENESYS exchange program in Nagano and Tokyo; selected among 250 students across India, 2011.